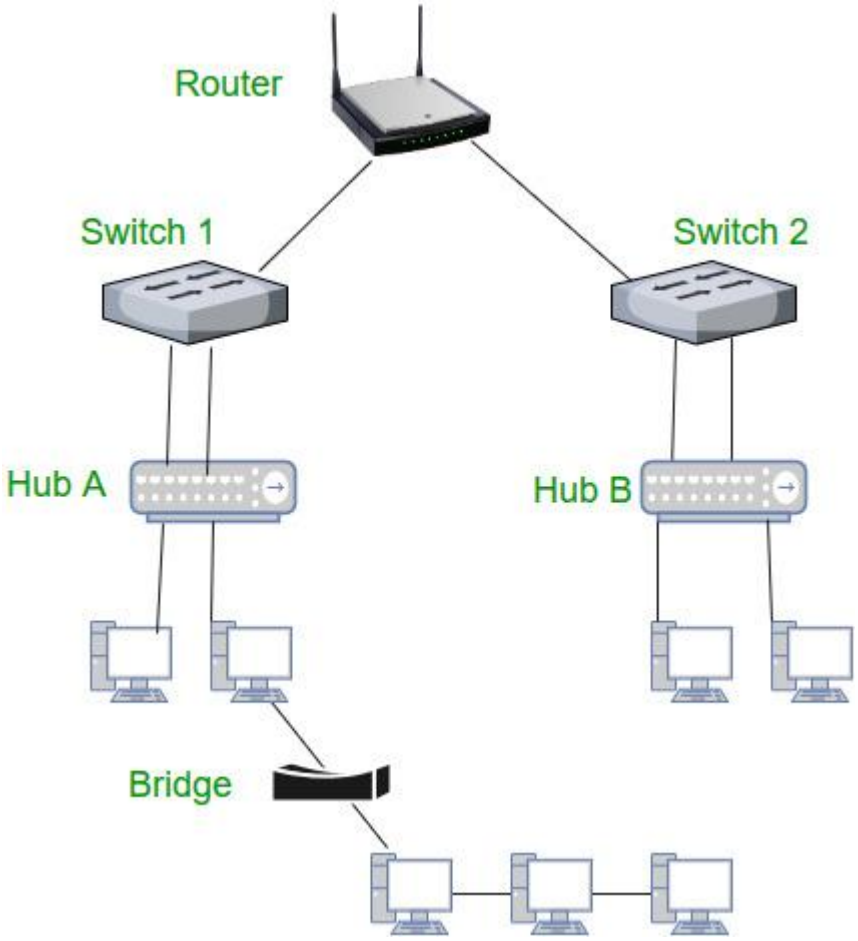
**Network Devices**

**1. Home Network Diagram**




**2. Comparison of Network Devices**

| **Device** | **Main Function** | **Real-World Use Case** |
|---|---|---|
| Router | Connects different networks and forwards IP packets. | Connects my home Wi-Fi network to the Internet so I can use Zomato or Instagram. |
| Switch | Connects devices within a LAN and forwards frames to the correct port. | Connects gaming PCs or office computers for local network communication. |
| Hub | Sends incoming data to all its other ports. | Can connect computers in a small, older LAN for basic communication. |
| Access Point (AP) | Provides wireless devices with access to a wired LAN. | Lets my smartphone connect to Wi-Fi to stream videos on YouTube. |
| | | |

**3. How Data Travels When I Order Food on Zomato**

1. I open Zomato on my smartphone and place a food order using Wi-Fi.
2. The access point provides the wireless connection between my phone and the home network. In a typical home, it is built into the Wi-Fi router.
3. The router forwards my phone's data toward the Internet through my Internet Service Provider (ISP).
4. Across the Internet, routers forward the data toward Zomato's servers. Switches may connect servers and other devices within data centers.
5. Zomato processes my order and sends a response back through the Internet and my router to my phone, confirming the order.

**4. Connecting a Gaming PC Far from the Wi-Fi Router**

Two possible solutions are:



Option 1: Powerline Adapter

- Uses the home's existing electrical wiring to carry network data.
- Connect one adapter to the router using Ethernet and plug it into a wall socket. Connect the second adapter near the PC and run an Ethernet cable from it to the PC.
- Useful when running a long Ethernet cable is inconvenient. Performance depends on the home's electrical wiring.



Option 2: Wi-Fi Extender with an Ethernet Port

- Receives the router's Wi-Fi signal and extends coverage.
- Place it where the Wi-Fi signal is still reasonably strong, then connect the gaming PC to its Ethernet port.
- Convenient when the PC has no Wi-Fi capability, although gaming latency and speed depend on the wireless connection.

My recommendation: Choose a powerline adapter if the electrical wiring is suitable and you want a wired connection. For competitive gaming, a direct Ethernet cable is usually the best option when practical.